

1 · The question, and what this project set out to test

The Platonic Representation Hypothesis (Huh et al., 2024) in one sentence, the project's target in one sentence, and the difference between them.

The hypothesis

Models trained separately — different data, different objectives, sometimes different senses entirely — are said to be converging on the same underlying picture of the world. The name comes from Plato: there is one reality, and each model sees a shadow of it.

A CAT ON A CHAIR

DINOv2 sees pixels $\rightarrow x_{\text{img}} \in \mathbb{R}^{768}$

bge-m3 reads text $\rightarrow x_{\text{txt}} \in \mathbb{R}^{1024}$

$\text{cosine}(x_{\text{img}}, x_{\text{txt}}) \approx 0.00 \leftarrow$ the coordinates share NOTHING

and yet, in BOTH spaces:

$\text{dist}(\text{cat-on-chair}, \text{cat-on-sofa}) < \text{dist}(\text{cat-on-chair}, \text{airplane})$

That gap is the whole project. The numbers are unrelated; the relations are not.

What this project tests

The hypothesis as usually stated is not directly testable — 'converging on the same picture' has no units. This project narrows it to something that can be measured, and then measures it:

CLAIM UNDER TEST

independently trained models share representations UP TO A LINEAR TRANSFORMATION

which means: there exists ONE matrix W , fitted in closed form, such that

$$x_A \cdot W \approx x_B$$

no training, no gradients, no fine-tuning — just a solve

Three things make that a real test rather than a slogan. It is **falsifiable**: if no linear map works, the claim dies. It is **bounded**: a linear map is the weakest useful transformation, so success cannot be explained away as a powerful network learning the task by itself. And it is **useful**: if it holds, components from different models become interchangeable, which is an engineering result and not only a scientific one.

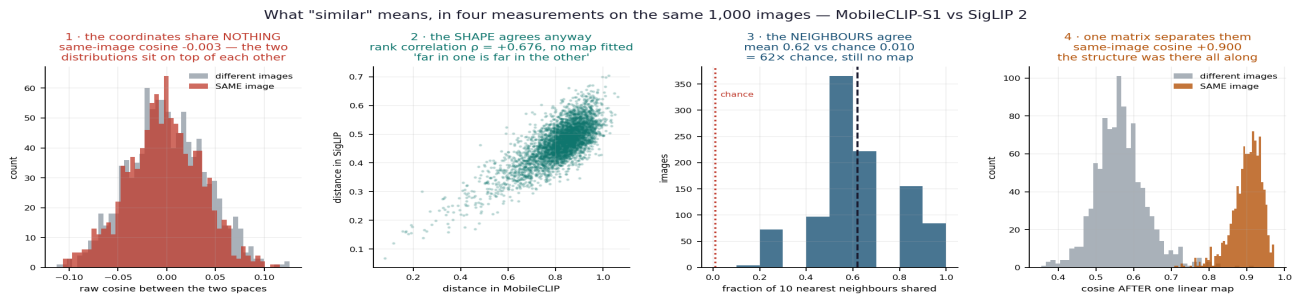
Two systems were built to test it

system	what it is	what it answers
Experiment B	two vision encoders, one 0.79 MB linear adapter, shipped as a Core ML artifact	Can two specific models be made interchangeable, well enough to deploy?
The Hub (Series G)	seven encoders, four training objectives, one 512-dimensional shared coordinate system	Can MANY models meet in one space, so a component trained on one serves the rest?

They answer different questions and carry different numbers. Keeping them separate matters: Experiment B's headline figure does not belong to the hub, and vice versa.

2 · Five kinds of similarity, from weakest to strongest

"Are these two spaces similar?" is not one question. It is five, and they can have different answers for the same pair of models. Every claim in this project names which level it is about.



Panels 2 and 3 use NO fitted map. The agreement they show cannot have been created by the matrix in panel 4 — it was already there, waiting for coordinates.

The same 1,000 images, four measurements. Panels 2 and 3 use NO fitted map — the agreement they show was already there.

level	the question, in plain terms	for the cat on a chair	how it is measured	result here
1 · shared NEIGHBOURS	Do the same things sit nearby?	in both spaces, is cat-on-sofa among its nearest items?	k-NN overlap, no map fitted	YES, strongly — hundreds of times chance at small k
2 · shared ORDER	Do they agree on what is nearer than what?	is cat-on-chair closer to cat-on-sofa than to an airplane, in both?	Spearman correlation of pairwise distances, no map	YES, moderately — ρ 0.29 cross-modal, up to 0.76
3 · shared STRUCTURE	Does the whole similarity pattern match?	does every item relate to every other in the same way?	CKA / RSA	PARTLY — 6.3x a random control, but weak in absolute terms
4 · shared SUBSPACE	Can one be turned into the other by a linear map?	can the image vector be mapped onto its caption's direction?	ridge regression, then retrieval	YES — 88–95% of a jointly-trained ceiling
5 · same GEOMETRY	Are they the same shape, just rotated?	does a pure rotation align the two clouds?	orthogonal Procrustes	NO — 0.038 against ridge 0.592. Related, not rigid

Reading the ladder

Each level is a stronger claim than the one above it, and the project's central result is that the answer **changes as you climb**. Neighbours agree strongly. Order agrees moderately. Global structure agrees weakly. A linear map works well. A pure rotation fails.

what that pattern means

strong at level 1, weak at level 3

→ the models agree about NEIGHBOURHOODS, not about global arrangement

level 4 works but level 5 fails

→ the spaces are related by a STRETCH, not merely a turn

This is why the project's conclusion is narrower than the hypothesis it started from. Models do not build the same space. They build different **charts** of one shared structure — like two map projections of the Earth, which disagree about distance and area while both keeping Spain next to France.

3 · The measurements, and how to read them

Every metric is a question with a blind spot. This page is the reference: what each one asks, when it lies, and what a high or low value actually means.

metric	what it asks	when it MISLEADS	how to read it
cosine	do two vectors point the same way?	in a collapsed space every pair reads +0.999 — the number is high and meaningless	always check the space's mean pair-cosine first
R ² (held-out)	do the predicted numbers match the target's?	if the target has tiny variance, R ² goes hugely negative even when retrieval is fine	reconstruction quality WITH magnitude; distrust on collapsed targets
R@1 / R@5	does the right item come first out of N?	meaningless without N — 0.967 of 10 and 0.557 of 1,000 are not comparable	the deployed question. Always quote against chance 1/N
% of ceiling	how much of a jointly-trained system survives?	if the ceiling was measured under a different protocol	the honest cross-experiment yardstick
k-NN overlap	are my neighbours your neighbours?	the raw fraction rises with k simply because chance does	read the RATIO to chance (k/N), never the raw number
CKA	do the two similarity structures match?	wider embeddings score higher even when independent	compare the RATIO to a random control, not the absolute
Procrustes	is it just a rotation?	—	low against a high ridge means anisotropy: a diagnosis, not a failure
effective rank	how many directions carry information?	—	below ~10% of width means collapsed; predicts almost everything else
pair-cosine	how alike are two RANDOM items?	—	near 0 healthy, +0.999 collapsed. The fastest single diagnostic
verification AUC	same item, or a random impostor?	the negatives are easy, so scores sit near 1	0.999 here means the QUESTION was easy, not that the model is perfect
rows per dimension	is the fit even powered?	below 5, a negative result may just be starvation	check BEFORE believing any failure

The lesson this table encodes

The project's recurring discovery is that **a metric is only interpretable relative to the geometry it is computed in**. The same system scored cosine 1.000 and retrieval 0.160 on the same data — not a contradiction, but two questions with different answers. Every surprising number in this project turned out to be a metric artifact or a geometry problem, never a broken pipeline.

4 · Experiment B — two encoders, one matrix, 0.79 MB

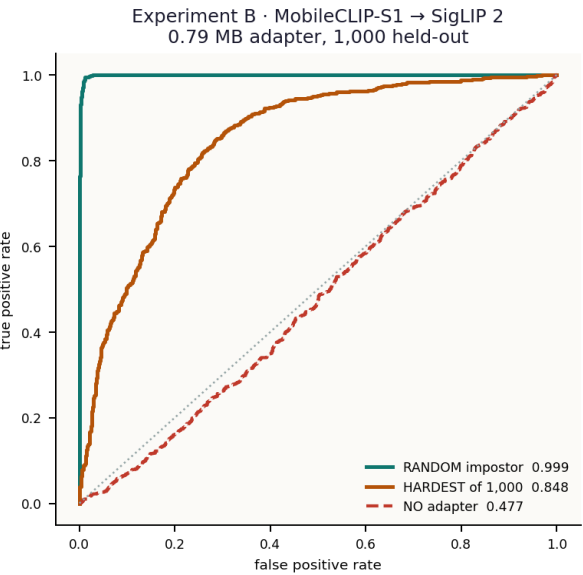
The narrowest version of the claim, tested to deployment. A phone-sized vision encoder and a server-sized one, never trained together, made interchangeable by a single linear map.

THE SETUP

MobileCLIP-S1 (phone tier) → 512 numbers per image
SigLIP 2 (server tier) → 768 numbers per image

THE FIT — closed form, no training
 $W = (X^T X + \alpha I)^{-1} X^T Y$ on 3,000 image pairs

THE TEST — on 1,000 images the fit never saw
phone vector · W → does it retrieve the SAME photo in the server index?



Why AUC 0.999 is not overfitting

not overfitting — held out
fitted on 3,000 pairs, scored on 1,000 unseen. Overfitting would inflate every score; it does not.

the same map, three difficulties
vs a random impostor 0.999
vs the hardest of 1,000 0.848
single-best R@1 0.987
A memorised map does not fall 0.15 on harder negatives.

0.999 = an easy question
"same image, or a random other one?" — usually a dog against a kitchen. Near-ceiling is expected here.

quote all three, never 0.999 alone
0.999 typical impostor · 0.848 hardest negative · 0.987 one-of-a-thousand. One system, three questions.

control
remove the adapter: AUC 0.477, chance. Everything above is the 0.79 MB matrix.

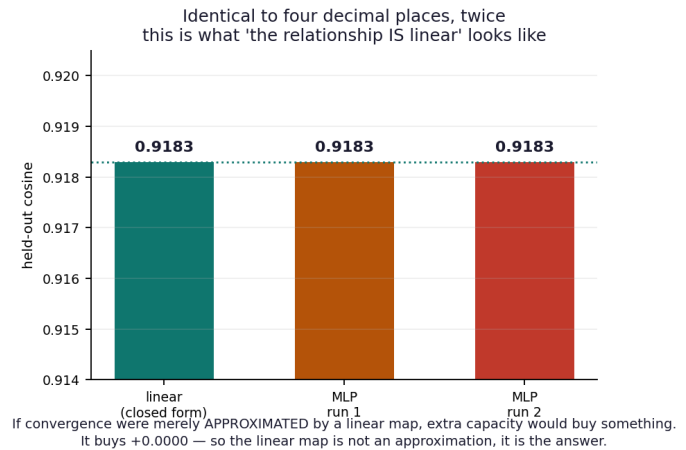
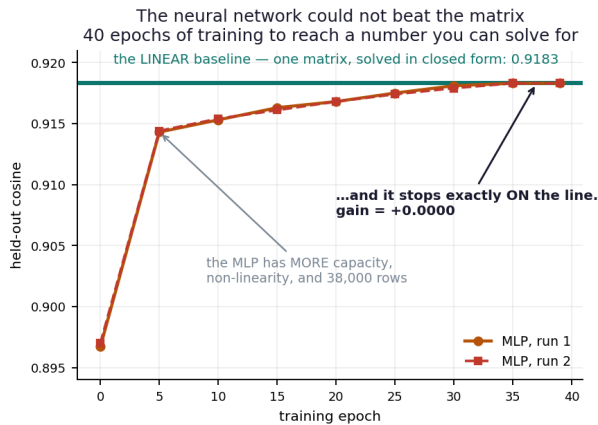
The same adapter, asked three questions of increasing difficulty. The spread is what proves it is not memorisation.

measurement	result	what it means
raw, no adapter	R@1 0.001 = exact chance	the two spaces share nothing usable in their own coordinates
ridge adapter	R ² 0.592, cosine 0.900	one matrix recovers most of the relationship
MLP instead of a matrix	the network FAILED to beat the matrix — gain +0.0000	given more capacity, non-linearity and 38,000 rows, 40 epochs of training arrived at exactly the number a closed-form solve produces. Not 'a linear map is good enough' — the relationship IS linear
retrieval	R@5 95.3% of a jointly-trained ceiling	good enough to deploy
Procrustes (rotation only)	R ² 0.038 vs ridge 0.592	a pure rotation fails — the spaces differ in stretch as well as orientation
shipped artifact	0.79 MB, fp16, iOS 16+	a phone can hold the entire translation between two model families

5 · The strongest evidence in the project is a failure

Finding 1 is usually stated as "convergence is linear". That undersells it. The result is that a neural network, given every advantage, could not beat a matrix.

The contest. A neural network was given every advantage over the matrix — non-linearity, far more parameters, twelve times more training data, forty epochs of gradient descent — and it could not beat it. It did not come close and fall short; it landed on exactly the same number, 0.9183, to four decimal places, on two independent runs.



Real training traces from both runs. The teal line is what one matrix achieves with no training at all.

Why that failure is the strongest evidence in the project: if convergence were merely APPROXIMATED by a linear map, extra capacity would buy something — a little, or a lot, but something. It bought +0.0000. A linear map is therefore not a convenient approximation of the relationship between these two models. It is the relationship.

6 · The Hub — seven experts, one meeting place

Experiment B connects two models. Real systems have many. Connecting them pairwise does not scale; a shared coordinate system does.

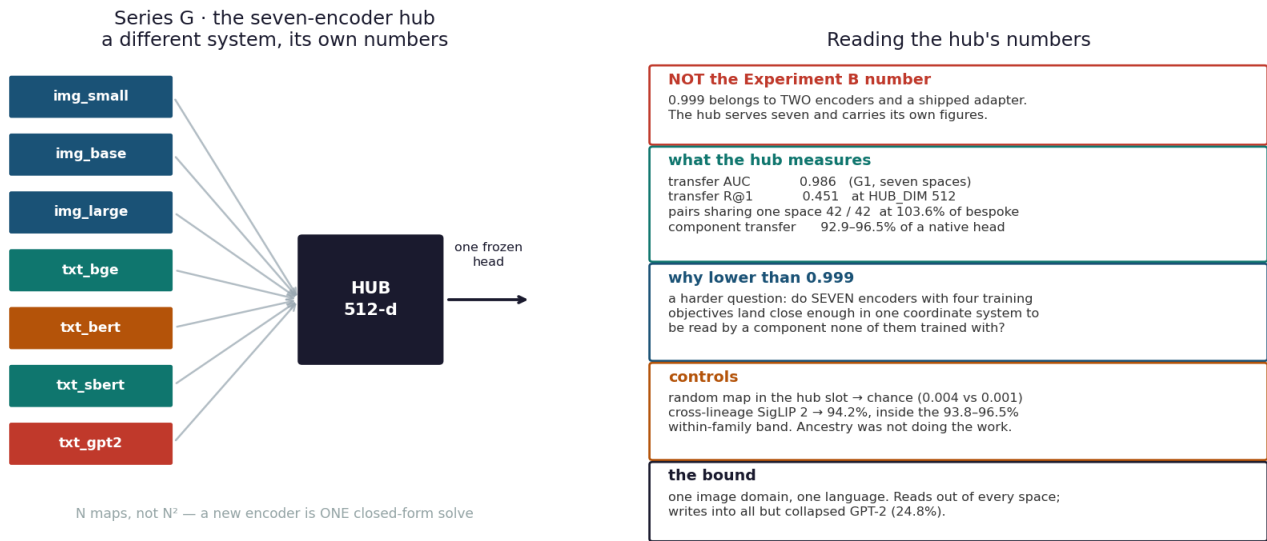
THE PROBLEM WITH PAIRS

7 encoders, connected pairwise → $7 \times 6 = 42$ maps
add one more model → 14 MORE maps to fit

THE HUB

7 encoders, each with one map INTO a shared space → **7 maps**
add one more model → **1 map**, and nothing else changes

N maps instead of $N(N-1)$



The hub is not a trained network — it is a whitened PCA plus one ridge solve per encoder. Nothing to maintain, nothing to retrain.

Why many small experts beat one large generalist

property	one large model	seven experts + hub
adding a capability	retrain or fine-tune the whole model	fit ONE new map; every existing map untouched
replacing a component	risky — everything depends on it	swap one arm, refit its map alone
cost of an update	full training run	a closed-form solve, seconds
specialisation	one model averaged over all domains	each expert strong in its own domain
measured cost of sharing	—	103.6% of hubs built for each pair alone — sharing is FREE
measured benefit	—	a component trained on ONE encoder works on encoders it never saw, at 92.9–96.5% of native

The honest cost, since closed. Hub width looked like a trade-off: transfer rises to 0.489 at 512 dimensions and falls to 0.312 at 768. It is not one. The cliff is least squares dividing by a near-null singular value, which appears exactly when hub width reaches the source encoder's rank - a numerical artifact of the fit, not a property of shared coordinates. The rule is $\text{hub_width} < \text{dim}(\text{source encoder})$, and a truncated or regularised head lifts the constraint entirely (report, E.8).

One GPU pass, then a ruler. The only step that needs a GPU is the one-time encoding pass that builds the caches · the eight encoders run forward over 9,533 items once, on Colab (T4/L4). Everything after is closed-form linear algebra on CPU: the ridge maps, the hub, all transfer. The expensive hardware makes the raw material once; the science runs without it · the same economy as the 0.79 MB adapter that ships without a GPU.

7 · The cave — what the allegory does and does not license

Plato's prisoners see shadows on a wall and take them for reality. The analogy is genuinely useful here, and it is also the place where this project's claim is most easily overstated.



Left: the measured version, two encoders. Middle: seven encoders positioned by how much they agree, with no map fitted. Right: the mapping, including what it does not license.

Plato	this project	the cat on a chair
the FORMS — one reality	the structure of the world that all encoders are trained on	the actual fact that cats sit on chairs, and that this resembles cats on sofas
the PRISONERS	seven encoders, trained apart, never compared notes	DINOv2 saw the photo. bge read the caption. Neither saw the other.
the SHADOWS	each encoder's raw vectors	x_img and x_txt: faithful casts of the same fact, privately rotated
the TURNING	one linear map per encoder	not new sight — the head turned so both shadows line up
walking out	this is where the analogy STOPS	no encoder ever sees the form directly. They agree with each other; that is all we measured

Where the allegory misleads

Plato's prisoner turns and sees **one** object casting **one** shadow. That would predict that two encoders agree about everything, at every scale. They do not. Agreement is strong for near neighbours and fades to almost nothing across the whole space — so the honest picture is not one object and one shadow, but **two different projections of the same territory**, each faithful locally and distorted globally.

the claim the evidence REFUTES

"good models converge to the same embedding space"

→ raw cosine ≈ 0 , global agreement $\approx$ chance

the claim the evidence SUPPORTS

"models are different CHARTS of one shared relational structure"

→ coordinates differ entirely; local relations survive the change

8 · What was found, proved, and discovered

Nine findings, each held-out and each quoted against a control, plus the two discoveries that were not being looked for.

#	finding	the evidence
1	Convergence is linear — proved by a failure	a neural network with more capacity, non-linearity and 38,000 training rows COULD NOT BEAT a closed-form matrix. Not narrowly: 0.9183 against 0.9183, gain +0.0000 , on two independent runs. Extra capacity buys nothing, so the linear map is not an approximation of the relationship — it IS the relationship
2	Related, not rigid	a rotation-only map scores 0.038 where a full linear map scores 0.592 — a 15× gap. The map's singular values span 69×, so the spaces differ in STRETCH along different directions, not merely in orientation
3	Deployable	95% of a jointly-trained ceiling from a 0.79 MB fp16 matrix; verification AUC 0.999
4	It crosses modality	a vision encoder that never saw a word of text reaches 40.9–58.3% of a jointly-trained ceiling inside a text space — up to 358× chance . Nothing in either training run rewarded agreement with the other
5	It grows with capacity	+15.3 points of ceiling per decade of parameters, six monotone measures, over a 13.8× range
6	Isotropy, not similarity training	whitening a raw language model's space — label-free, linear — reaches 102.6% of an encoder trained contrastively on 1.2 billion pairs
7	The correspondence identifies itself	Gromov-Wasserstein recovers 39.7% of pairings exactly from shape alone, 595× chance, with zero paired examples
8	One shared space, actually used	42 of 42 encoder pairs share one 512-d hub at 103.6% of bespoke; a head transfers to unseen encoders at 92.9–96.5%, including an unrelated lineage at 94.2%
9	The mechanism predicts	a pre-registered prediction about two unseen encoders confirmed at rank correlation +1.00

Two things discovered rather than sought

discovery	what it says	why it matters
The sharing is LOCAL	neighbourhood agreement runs hundreds of times chance; global agreement falls to about twice chance. Confirmed on three separate metrics, including the one the original authors designed	it narrows the Platonic claim to its defensible form, and it explains why a linear map reaches most but not all of the ceiling
Isotropy has two opposite effects	the same geometric property HELPS reading a known correspondence and HURTS finding an unknown one - measured twice. The cliff was once a third instance; it is rank, not isotropy (E.8)	one property, opposite prescriptions: there is no single 'good geometry', only geometry good for a stated task

And the methodological result. Ten explanations offered during this work were tested and recorded as false - two of them the project's own, and one entered on a faulty test and formally withdrawn - rather than quietly dropped. A claim that survived a test which could have killed it is worth more than one never exposed to one.

9 · Caveats, and what could be done next

Every claim above has a boundary. Naming them is not modesty — an unbounded claim cannot be tested, and a reviewer will find the boundary anyway.

What the results do NOT say

claim	measured scope	NOT claimed
convergence is linear	GPT-2/Pythia; MobileCLIP/SigLIP; DINOv2 ×3; four text encoders	all architectures, all scales
alignment grows with capacity	DINOv2 family, 22M–304M parameters	behaviour beyond 304M, or in other families
isotropy rescue	one collapsed target, two image scales	that isotropy is SUFFICIENT — it is necessary only
pair-free recovery	39.7%; both encoders contrastive image-text	a deployable system that needs no pairs
the shared hub	7 transformer encoders, 4 objectives, COCO, English; held-out SigLIP 2 and ConvNeXt	a universal space for all models, domains or languages
writing into a space	reads out of all seven; writes into all but collapsed GPT-2 (24.8%)	that a collapsed space can be reconstructed for ranking
structure is shared	LOCALLY — neighbourhoods agree, global geometry diverges	global metric identity
domain	one image domain, one language, one caption corpus	open-world scale, other domains or languages

What could be done next, in order of value

next test	the question	cost	why it matters
A convolutional encoder	every encoder here was a transformer. RUN, and closed: ConvNeXt-base joins at 96.7% of native	ran - control at chance (0.001)	ABOVE the within-family band (93.8-96.5%); architecture does not bound the claim
Experts vs one large model	do K small domain experts in a shared hub match one big general encoder, at matched parameters?	medium	turns the engineering argument into a benchmark someone outside this field would care about
More encoders per group	shared OBJECTIVE predicts agreement better than shared lineage — but on as few as one pair per cell	medium	the most interesting causal hint in the project, and currently the least powered
Other languages and domains	does any of this survive outside COCO captions in English?	high	the broadest of the stated limits
Non-linear hubs	a neighbourhood-preserving embedding might carry local structure where whitened PCA does not	medium	ruled out here by the no-training constraint, but it is the natural sequel

Reverse-direction transfer	does the hub carry information recoverable TOWARD an image space, not just FROM one? Source/target roles are independent (G10, SBERT) - untested for images	medium first image-target test; ConvNeXt is the cleanest case since it was never a hub member, so no overlap question to declare. Mirrors G10's source-composition sweep, applied to which encoder trains the head
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Scale the text side	PRH's own capacity ladder predicts alignment rises with encoder size - tested here only up to bge/BERT/GPT-2 scale (max 1024-d). Does a 7B+ text encoder push transfer and shape agreement higher, as the ladder predicts?	medium the closed-form design means adding a larger encoder costs one entry-map solve, same as any other. The scale caveat an examiner is most likely to raise - answered in advance if this runs before the defense, conceded cleanly if not
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THE PROJECT IN ONE PARAGRAPH

Two models never share coordinates · raw agreement is exactly chance. But they describe the same world, and the translation between them is a single matrix small enough to fit on a phone. What they share most strongly is LOCAL: they agree about what is near what, and diverge about the shape of the whole. The coordinate frame that makes structure readable is cheap; the arrangement of meaning across directions is what training actually buys.